

Edge detection in Amateur Cricket by the use of Triaxial Accelerometers, Microphones and Machine Learning

Luke O'Dair supervised by Dr Andrew Gallant

Abstract—Caught behind is one of the most common methods of dismissal within cricket. However, it is also one of the hardest decisions to make, contributing to the tolerance within the International Cricket Council's (ICC's) required decision-making accuracy of 91% for professional umpires. In this paper, a purpose built device records the accelerations and sounds created during off drives to assist in the decision-making process, by use of supervised machine learning. Data is recorded using a 3-axis accelerometer and microphone attached to the back of the bat, before being transformed to the frequency domain using the Fast Fourier Transform (FFT). A variety of features of the FFT data were extracted and tested on different machine learning algorithms in order to classify impacts as either a hit, edge or miss. The most accurate algorithm was found to be a linear support vector machine (SVM) when trained on the covariance and standard deviation of the post impact accelerometer FFT data. The average accuracy of 87.8% and peak accuracy of 89.3% are comparable to the ICC requirements. Furthermore, the accuracy of the algorithms improve as the dataset increases in size, suggesting that the device could exceed the accuracy of umpires in the future.

Index Terms—Edge Detection, Machine Learning, Support Vector Machine

I. INTRODUCTION

Over the last decade, technology has had a dramatic impact on the way sport is officiated, coached and viewed. The introduction of a Decision Review System (DRS) in October 2009 to professional cricket [1] allowed players to challenge the decisions of on-field umpires by the use of various different technologies and an additional umpire. This aimed to prevent poor decision-making affecting the outcomes of matches. However, this technology has not filtered down into the amateur game. The research presented examines an alternative system designed to improve decision-making accuracy on caught decisions within amateur cricket, building on the research of Busch and James [2], Fallon et al. [3] and Ting and Chilukuri [4] by the incorporation of Machine Learning.

Machine learning (ML) is defined as a field concerned with constructing computer programs that automatically improve with experience [5]. Banko and Brill [6] found that as datasets increase in size, ML algorithms became more accurate with less variation between different algorithms performance. Thus concluding that the power of a ML algorithm is defined by the size of the dataset. The aim of this research is to reach the levels of accuracy in decision-making required of umpires by the International Cricket Council (ICC) of 91%

[7]. Failing this, the research will investigate the most effective combination of ML algorithms and modern technologies to classify impacts between a cricket bat and ball. Additionally considering whether algorithms become more accurate as the dataset increases in size, to assess whether the ICC tolerance could be exceeded in the future.

The combination of sports and technology has been used in a variety of similar research applications for classifying impacts. Cross [8] used piezoelectric disks to determine the sweet spot for hitting a tennis ball with a tennis racket. Fallon et al. [3] built on this work to predict the impact location of a baseball and bat given the data obtained from different technologies attached to the bat. The paper found that accelerometers and microphones were the most effective methods, both with prediction accuracies of 63.0% within 5cm of the impact location.

Within cricket itself, Sarkar et al. [9] conducted a similar study to that of Cross [8], but on cricket bats. Accelerometers were attached to the back of a bat and gloves in order to determine the bats sweet spot upon impact with a cricket ball. Busch and James [2] also used accelerometers mounted to a cricket bat to extract useful information about the batsman's swing. The relative magnitude of the acceleration in the Y-axis enabled the angle of bat and therefore shot type to be determined.

Ting and Chilukuri [4] attempted to design an intelligent decision-making system using pre-existing DRS data for caught decisions. The method presented could classify impacts with an accuracy of 80.0%, but was limited by its use of simple classifiers. Rock et al. [10] expanded on this research to design, build and test the infrastructure to utilise this system in game situations.

Within the private sector, there has also been progress made in this area. Dr Hawkins claimed in April 2016 that Hawk Eye were planning to release a chip to be attached to bats to assist in the decision-making process in club cricket [11]. However, no further details about this chip have been published.

This paper contributes to the research in this field in two ways. Firstly, it investigates the feasibility of mounting a device to a cricket bat to measure both acceleration and sound created through impact. Secondly, it investigates the effectiveness of ML in classifying impacts using the data obtained from the device to assist in the caught decision-making process, a concept that has not been considered before. Section II describes and summarises the criticism of the use of technology within professional cricket. Following this, the ML

theory behind the research and experimental method utilised are described in Section III and IV respectively. The results of the feature and ML algorithm testing for classification of impacts are presented and discussed in Section V.

II. BACKGROUND

The DRS functions with each team having a set number of challenges per innings. If a challenge does not lead to the decision being overturned; the team loses a challenge until it has none remaining. The technologies that are used as part of the system are summarized below [12], [13]:

- **Snickometer** (Figure 1) uses microphones in the stumps attached to an oscilloscope that are overlapped with real time images. This allows the third umpire to see when the sounds were created and if the batsmen hit the ball or not for caught decisions.



Fig. 1. Snickometer being used to make a caught decision against Kevin Pietersen [14]

- **HotSpot** (Figure 2) uses infra-red cameras to record the thermal energy transfer from the ball to the bat due to the friction created. This also allows the third umpire to see whether the ball made contact with the bat or not for caught decisions.



Fig. 2. Hotspot being used to make a caught decision [15]

- **Hawkeye** (Figure 3) uses six cameras to track the ball from when the bowler releases it to when it strikes the batsman. A 3D image is created and an algorithm is employed to predict the likely trajectory of the ball. This allows the third umpire to make decisions on leg before wicket (LBW) appeals.



Fig. 3. Hawkeye being used to make an LBW decision against Gary Ballance [16]

For the first time, the DRS has allowed a true evaluation of the performance of umpires within the game of cricket. Shivakumar [17] conducted a statistical analysis on the 1201 decisions challenged using the DRS between 2009 and 2014. It was found that 25.81% of decisions were overturned, which is significantly higher than the 9% tolerance stated by the ICC.

Collins and Evans [18] argue that the use of technology in sports has led to unfair scrutiny on officiating without acknowledging the tolerances within the technology. Borooah [1] also argues that the cost of the DRS is not worth the increase in accuracy in decision-making. Borooah estimates the cost for a 1% increase in decision accuracy is around £80,000 per test match and that improving umpires training would lead to a similar increase in accuracy for a fraction of the cost.

However, Collins [18] and Borooah [1] both conclude that the use of technology within sports is inevitable given the rapid rise of computing power and enhancements that it can provide to the viewing and playing experience. Despite this assessment, an equivalent system to the DRS has never been devised in amateur cricket.

One of the most common umpiring decisions within cricket is whether a batsman is out caught, accounting for 57.0% of all dismissals [19]. The most difficult decisions of this type are detecting when a batsman has very faintly hit the ball and is caught-behind by the wicketkeeper or slips. This is known as an edge and accounts for 16.3% of dismissals [19]. The research presented aims to investigate the ability of technology to differentiate between hits, edges and misses to assist umpires in making caught decisions.

III. THEORY

The method of ML that this research employs is supervised learning. This is where an algorithm or classifier is fed a large dataset of known examples and then extrapolates the trend from the variables or features provided for future predictions. Within supervised learning, there are a wide range of algorithms to choose from in terms of methodology and complexity. The different types of supervised learning algorithms used as part of this research are summarised below:

- 1) **Decision Tree:** creates a predictive model that maps input variables to conclusions about the items classification. Leaves represent class variables and branches

represent conjunctions of features that lead to classification. The number of branches defines the complexity of the algorithm [20], [21].

- 2) **Discriminant Analysis (DA)** is a method of ML based on the work of Fisher [22] that uses similar probabilistic principles to logistic regression to classify data. The complexity of the model can be varied to include hyperbolic, parabolic and elliptical functions [23].
- 3) **Support Vector Machines (SVMs)** are discriminative learning algorithms that separate classes using an optimal separating hyperplane. This hyperplane is then used to classify samples in the future. The kernel employed dictates the complexity of the hyperplane produced [24], [25].
- 4) **K Nearest Neighbor (KNN)** is an instance based learning algorithm that compares the similarity of a test sample to training examples by use of the Euclidean distance. The method of comparing the test sample to the training examples in its vicinity defines the complexity of the algorithm [26], [27].
- 5) **Ensemble** algorithms are comprised of multiple weaker models that are combined to make the overall prediction. This type of ML algorithm is becoming very popular as different models and methods to combine them are considered [28], [29].

This paper uses the Matlab Classification Learner Toolbox [30] to apply different supervised ML algorithms, which are summarized by type in Table I.

The method by which the accuracy of a supervised learning classifier is determined is known as validation. The holdout method [31] is the simplest form of validation, where the dataset with known classification is split randomly into a training and test set. The training set is used to train the ML algorithm, which then makes predictions on the test set. The accuracy of the ML algorithm is found by comparing the classification predictions to the known classification of the test set. The K-fold cross validation method [31] extends this method by splitting the known classification dataset randomly into K equal subsets. Each subset is utilised as the test set independently, with the other (K-1) subsets being used to train the ML algorithm. The holdout method is therefore repeated K times and overall accuracy of the classifier is determined as the average accuracy of the K classification predictions for each of the test sets. The research presented employs five fold validation throughout experimentation as it is appropriate for the dataset size [30]. This prevents overfitting of algorithms, excessive computational demand and facilitates objective evaluation of different algorithms accuracy.

IV. EXPERIMENTAL METHOD

A. Device Design

From previous research [3], the technologies chosen to classify the impacts were an accelerometer and a microphone. However, these devices would have to be positioned in a location where an impact with the ball would be improbable.

TABLE I
SUMMARY OF THE DIFFERENT MACHINE LEARNING ALGORITHMS
UTILISED AS PART OF THIS RESEARCH

Algorithm	Description
Simple Tree	Tree with up to 4 decision splits
Medium Tree	Tree with up to 20 decision splits
Fine Tree	Tree with up to 100 decision splits
Linear Discriminant	DA with a linear boundary between features
Quadratic Discriminant	DA with a elliptic, parabolic or hyperbolic boundary between features
Linear SVM	SVM with a linear kernel
Quadratic SVM	SVM with a quadratic kernel
Cubic SVM	SVM with a cubic kernel
Fine Gaussian SVM	SVM with a Gaussian kernel equal to $\sqrt{P/4}$ where P=no. of predictors
Medium Gaussian SVM	SVM with a Gaussian kernel equal to $\sqrt{P}$ where P=no. of predictors
Coarse Gaussian SVM	SVM with a Gaussian kernel equal to $\sqrt{4*P}$ where P=no. of predictors
Fine KNN	Number of neighbours set to 1
Medium KNN	Number of neighbours set to 10
Coarse KNN	Number of neighbours set to 100
Cosine KNN	Cosine distance metric used
Cubic KNN	Cubic distance metric used
Weighted KNN	Weighted distance metric used
Boosted Trees	Combination of medium decision trees through the Adaboost algorithm
Bagged Trees	Bootstrap aggregate combination of multiple fine decision trees
Subspace Discriminant	Combination of discriminant classifiers by use of the random subspace algorithm
RUS Boosted Trees	Combination of multiple trees, useful for skewed datasets

The devices were therefore mounted to the back of the bat. Additionally, a battery, microcontroller and switch would have to be mounted to power and control these devices. The design should aim to minimize the impact on the weight of the bat and not affect the propagation of vibrations through the bat, while also being robust enough to withstand multiple impacts.

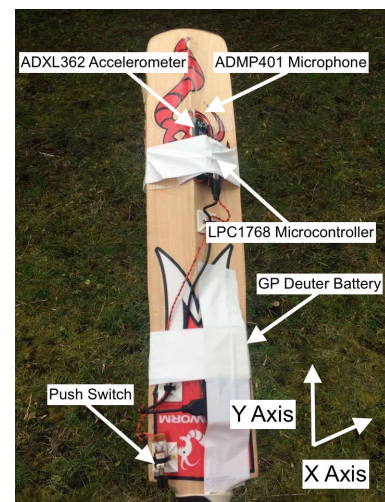


Fig. 4. Photo of the back of the test bat with the components and axes labelled. The Z axis is orientated out of the page perpendicular to the X and Y axes shown.

After a number of design iterations, the final design for the device can be seen in Figure 4. The back of cricket bats are not flat, hence the unconventional orientation of the X,Y and Z axes used. An Mbed NXP LPC1768 [32] microcontroller was chosen to obtain and store the data. Connected to this was a GP DEUTER 1800mAh battery [33] to power the device and a push switch to start and stop the readings. To record the vibrations in the bat, an ADXL362 3-axis accelerometer was utilised that reads to a precision of $(\pm) 8g (m/s^2)$ [34]. An ADMP401 microphone was chosen to record the sound of the sample with a signal to noise ratio of 62 dBA, a sensitivity of -42dBV and a maximum acoustic input of 120 dB [35]. A sampling frequency of 100 Hz was chosen for the measurements as it gave a compromise between storage constraints of the microcontroller and precision of the results. Code was written in C to control the components and store multiple readings on the mbed device.

B. Experimental Procedure

The experimental procedure was constrained to one specific cricket shot: the off drive. This was chosen as it is the most common shot by which batsmen are caught-behind [19]. To achieve this limitation, a method was required to feed balls at a consistent velocity, trajectory and position. This was achieved by using a BOLA professional bowling machine and BOLA practice balls [36] as can be seen in Figure 5. The speed chosen was 65 mph and location of desired bounce was a full delivery just outside off stump. In order to keep the results comparable, the same test subject was used for all readings. To achieve a consistent bounce, the research was conducted using specially designed surface to simulate a cricket pitch surrounded by nets to prevent the balls travelling any significant distance, as can be seen in Figure 6.



Fig. 5. BOLA Professional Bowling Machine used to feed the balls at a constant velocity, trajectory and position



Fig. 6. Nets to stop the balls post impact and mat to simulate a cricket pitch

The test subject would start the measurement by pressing the push switch located on the bat. The assistant would then raise the ball above the BOLA bowling machine, before feeding the ball into the machine. The ball would reach the batsman at which point he would play an off drive. The reading was stopped a few seconds after the ball had been

hit, edged or passed the bat. The result was then classified as either a hit, edge or miss and the process repeated. After 10 readings, the data was transferred from the LPC1768 to a computer for analysis and storage. Data processing was conducted using Matlab Student edition 2015b once the data had been transferred to a computer.

The definitions of a hit, edge and miss were established as:

- **Hit:** ball impacts with the middle of the bat and changes trajectory in the direction the batsman intended.
- **Edge:** ball impacts with the outside edge of the bat and changes trajectory not in the direction the batsman intended in a way so that the batsman could be caught-behind.
- **Miss:** ball does not impact with the bat despite the batsman attempting to hit the ball.

V. RESULTS AND DISCUSSION

A. Raw Data

In total, 206 samples were collected that can be separated into 132 hits, 47 misses and 27 edges. Figure 7 shows the raw data for one hit, edge and miss from the accelerometer and microphone respectively with the location of the impact or miss labeled. The impact between bat and ball for the hit and edge occurs at the peak in the sound recording. Similarly, the peak in the miss sound recording occurs when the ball hits the back of the net after passing the bat. On all three samples, this peak occurs after a sharp negative gradient in the Y-axis accelerometer reading. This is caused by the batsmen swinging the bat downwards in the Y plane, in order to make contact with the ball.

Inspection of the accelerometer data for the example hit show a clear similarity with the results found by Busch and James [2]. One major difference between the results is the oscillations in the X and Z axes prior to the impact, clearly seen in Figure 7. This was investigated using a slow motion camera during testing and was found to be caused by the individual batting technique of the test subject. The example hit and edge microphone data also show similarities to those collected by Rock et al. [10] in respect of the change of amplitude of the waveforms for the different types of impact. However, the miss sound waveform differs significantly due to the peak post miss. This is caused by the impact between the ball and the back of the nets during testing, unlike in a game situation.

The raw data obtained from the accelerometer and microphone was transformed into the frequency domain using the Fast Fourier Transform (FFT). This enabled the accelerations and sounds recorded to be compared and classified by extracting a variety of features from the FFT data. A summary of the features extracted from the accelerometer and microphone FFT data throughout experimentation can be seen in Table II, chosen from prior research [4], [10], [37], [38]:

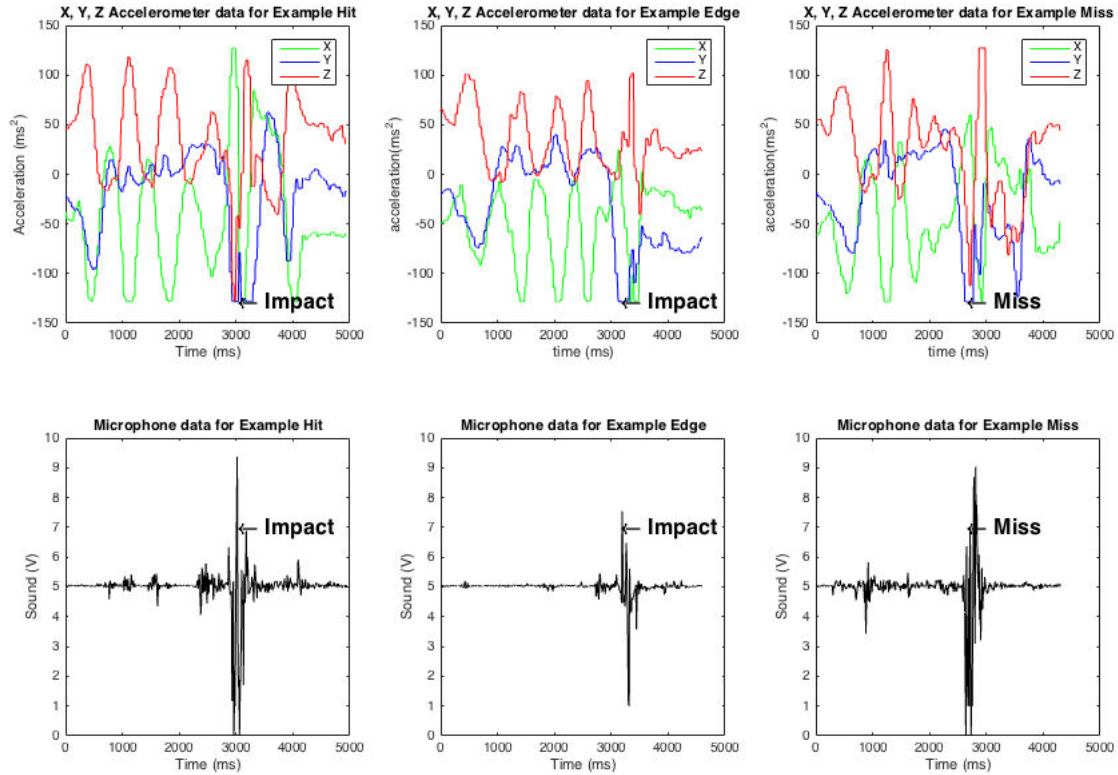


Fig. 7. Top row shows the raw data from the accelerometer for a hit, edge and miss moving from right to left in the X,Y and Z axis respectively. Bottom row shows the raw data from the microphone for the above samples. The location of impact/miss has also been labelled.

TABLE II

SUMMARY OF THE DIFFERENT FEATURES EXTRACTED FROM THE FFT OF BOTH THE ACCELEROMETER AND MICROPHONE DATA

Features	Description
1-4	Height, Location, Width and Prominence of largest peak in FFT of X Axis Accelerometer data
5-8	Height, Location, Width and Prominence of largest peak in FFT of Y Axis Accelerometer data
9-12	Height, Location, Width and Prominence of largest peak in FFT of Z Axis Accelerometer data
13-15	Average of FFT of Accelerometer data for X,Y, Z Axes
16-18	Standard Deviation of FFT of Accelerometer data for X,Y, Z Axes
19-21	Covariance of FFT of X and Y Axis Accelerometer data
22-24	Covariance of FFT of X and Z Axis Accelerometer data
25-27	Covariance of FFT of Y and Z Axis Accelerometer data
28-31	Height, Location, Width and Prominence of largest peak in FFT of Microphone data
32	Average of FFT of Microphone data
33	Standard Deviation of FFT of Microphone data

B. Accelerometer Readings Feature Selction

In order to remove the noise in the accelerometer reading, the pre and post impact sections of the data are separated using a function that finds the maximum negative gradient in the Y axis accelerometer readings within Matlab. The FFT of the post impact and full accelerometer readings for the same three results in Figure 7 can be seen in Figure 8. The frequency

components displayed are between 0-10Hz as this is where the only significant frequency components were identified. By inspecting multiple results, it became clear the post impact FFTs have a lower number of prominent frequencies but comparable amplitudes to the full sample FFTs.

Experimentation was conducted to find features from both of the accelerometer readings FFTs that when fed into ML algorithms, most accurately classify impacts. Each accelerometer feature from Table II was extracted from both the full and post impact FFT accelerometer data, before being tested independently on four different ML algorithms: Linear SVM, Linear Discriminant, Simple Tree and Medium KNN. These algorithms were chosen to reduce computational demand and ensuring each type of algorithm available was represented (excluding ensemble). The algorithms were tested using three classes (hit, edge and miss) with five fold validation on the entire dataset of 206 samples. The aim of this stage of testing was to assess whether the isolation of the impact and vibrations in the bat had any impact on the accuracy of classification. The average accuracy across the four algorithms for the features extracted from the full and post impact FFTs, are shown in Figure 9.

It is clear from these results that the post impact accelerometer features result on average in more accurate ML algorithms than the full signal accelerometer features. In addition, it is

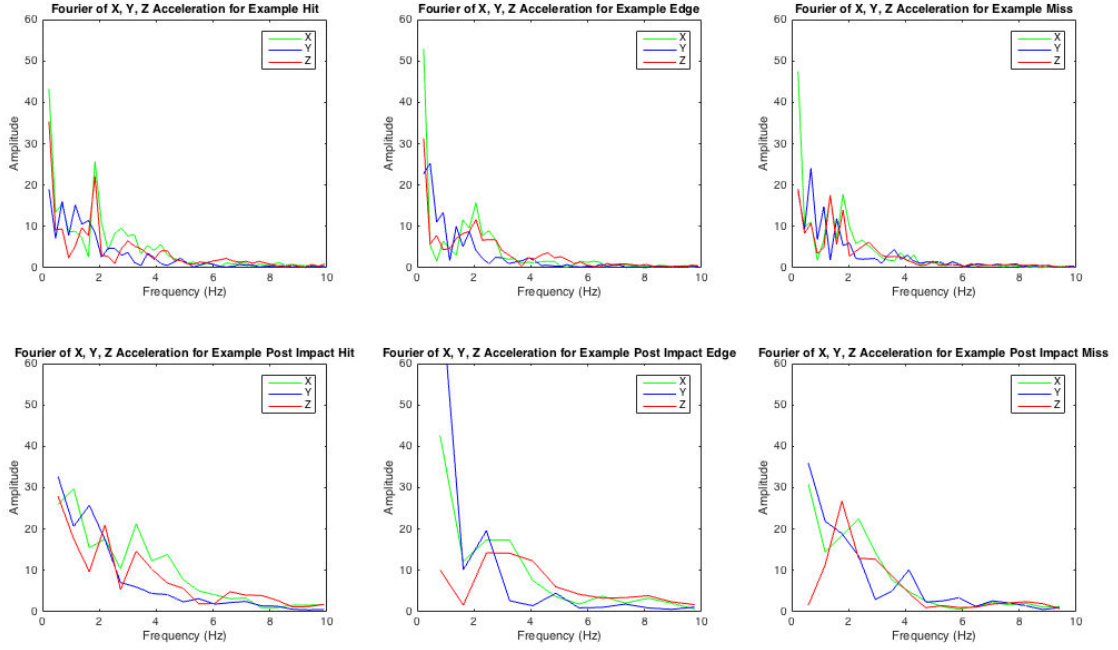


Fig. 8. FFT of the X, Y and Z axis accelerometer example data in Figure 7 for the whole sample (top) and post impact sample (bottom) from 0-10Hz.

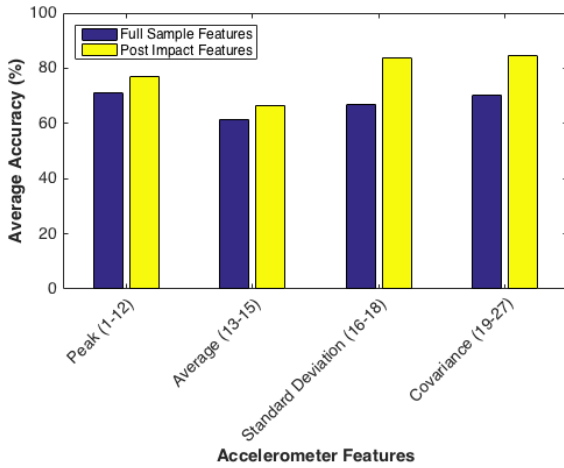


Fig. 9. Average accuracy of the features extracted (as described in Table II) from the full sample FFT and post impact FFT accelerometer data across four different ML algorithms (Linear SVM, Linear Discriminant, Simple Tree and Medium KNN)

clear that the standard deviation and the covariance features show a dramatic increase in accuracy compared to the peak and average features due to this effect. The comparatively small increase in accuracy for the average feature (13-15 from Table II) can be explained by inspection of the FFTs in Figure 8. The amplitudes within the samples FFTs for the post impact and full sample are similar and therefore average values will differ to a lesser extent. Furthermore, a similar effect for the peak

feature (1-12 from Table II) suggests that the largest peaks in both the post impact and full sample FFTs are being caused by the impact between bat and ball. The observation in the standard deviation feature (16-18 in Table II) suggests that the post impact FFTs become significantly more consistent. This could be caused by the removal of inconsistencies in the test subject's backswing prior to impact. In order to explain the accuracy increase in the covariance feature (19-27 in Table II), further testing was conducted. By breaking down the feature further into the relationships between the X, Y and Z axis FFTs respectively, it was found that the increase in accuracy was observed on all three interactions but most significantly on the covariance between the Y and Z axis FFTs. This subset of a feature showed an average accuracy increase of 18.3% in comparison to 7.2% and 6.4% respectively for the XZ and XY FFT covariance. This can be explained in the original accelerometer readings in Figure 7. The Y and Z accelerometer readings are a mirror image about the line ($a=0 \text{ m/s}^2$) prior to impact, but differ significantly post impact. Therefore, isolation of the impact will significantly affect this specific covariance feature.

In summary, the features extracted from the post impact accelerometer FFTs result in more accurate ML algorithms by an average of 10.5% than the features extracted from the full sample accelerometer FFTs. Therefore, the full sample accelerometer data was not used in further experimentation.

C. Microphone Readings Feature Selction

A similar experimentation procedure was conducted to find the most accurate features to describe the microphone

readings. From Figure 7, it is clear that there is significantly less noise in the microphone data than the accelerometer data. Therefore, no isolation of impact was required. The FFT of the microphone data for the same three results as in Figure 7 can be seen in Figure 10, plotted from 0-50Hz as this is where the significant frequency components were identified.

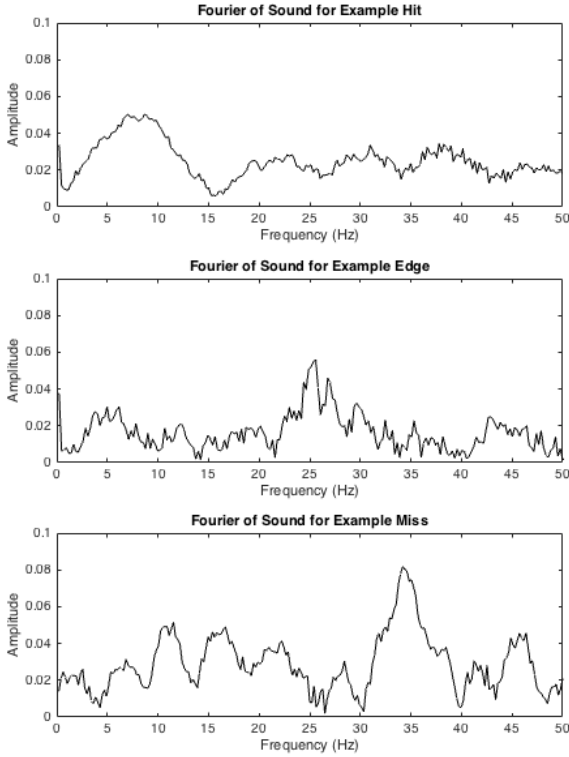


Fig. 10. FFT of the example microphone data in Figure 7 with peaks identified from 0-50Hz

The FFT of the sound readings have lower amplitudes of peaks but at higher frequencies than the FFT of the accelerometer readings. This is because the sound occurs over a much shorter time period than the vibrations in the bat and therefore appear less significant when transformed over the entire sample. The same test procedure as for the accelerometer features (see Subsection V-B) was utilised to evaluate the average accuracy of the features of the microphone reading FFTs described in Table II.

Figure 11 shows the results of the average accuracy of the four ML algorithms for the different features of the microphone readings FFTs. The sound features are less accurate at classifying impacts by 3% and 13.5% on average compared to the full and post impact accelerometer FFT features. From multiple sample observations, it became clear that the FFTs of the microphone readings for the different impacts were extremely consistent with fewer discernable patterns in the shapes of the waveforms, compared to the FFTs of the accelerometer data. This could explain why the Average and Standard Deviation features of the FFT (32 and

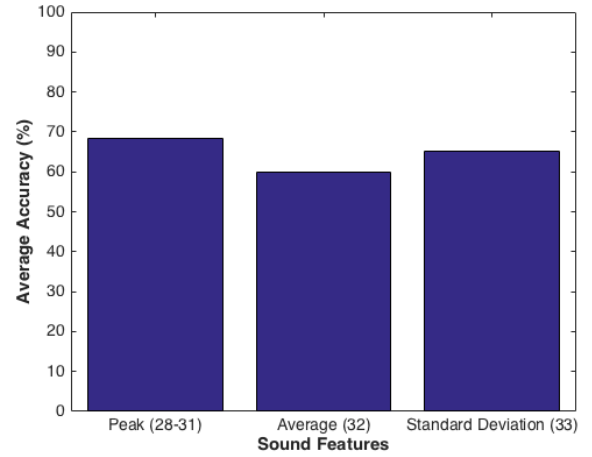


Fig. 11. Average accuracy of the features extracted (as described in Table II) from the microphone data across four different ML algorithms (Linear SVM, Linear Discriminant, Simple Tree and Medium KNN)

33 from Table II) were not as effective at classifying the impacts. However, the peak feature (28-31 from Table II) does perform comparatively to the features extracted from the accelerometer FFT. This suggests that this feature is extracting useful information about the frequency of sounds created during different types of impacts.

D. Combination of Features

To determine the most accurate combination of features from the accelerometer and microphone reading FFTs another experiment was conducted. The same sample size, validation procedure and number of classes were used, however all the ML algorithm's accuracy from Table I were computed. Features were added to the dataset in order of average accuracy from the previous stage of testing (excluding the full sample accelerometer FFT results). The aim of this experimentation was to determine whether the least significant features had an impact on overall accuracy of the algorithms, as removing them would lead to a computational time saving in training the algorithm.

Figure 12 shows the results of this testing presented as an average accuracy for each type of algorithm and an overall average accuracy across all the ML algorithms. It clearly shows that more features do not necessarily lead to a more accurate algorithm. In general as the number of features increase, the average accuracy of the algorithms decrease when employing this method for adding features to the model from prior testing. Discriminant algorithms were the most consistent in terms of average accuracy as the number of features increased. This is because these algorithms are probability-based and very effective at utilising the most and ignoring the least significant features within the dataset [23]. On the other hand, Tree algorithms were the least consistent as the number of features increased. This is because this type of algorithm requires parameter tuning prior to training [20]. From the overall average accuracy line in Figure 12, it is clear that the

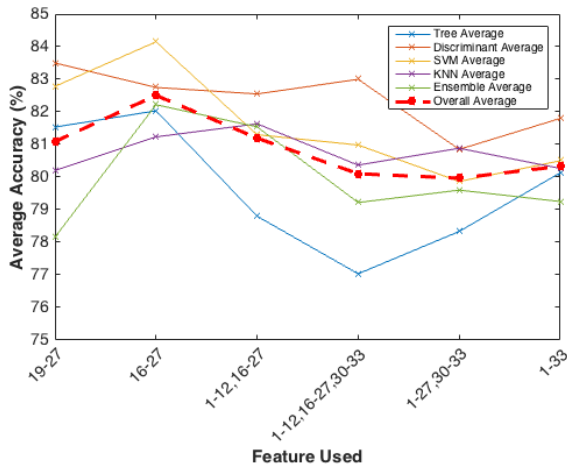


Fig. 12. Average accuracy of ML algorithms by type as number of features are increased in order of accuracy from prior testing in Subsection V-B and V-C. Results for each type of algorithm and overall average are joined by a line to demonstrate the trends as the features are added, rather than for interpolation

most accurate features for classifying impacts were the covariance and standard deviation of the post impact accelerometer FFT readings (16-27 from Table II). This result means that the microphone was not recording significant information to classify the impacts more accurately than the accelerometer on its own. This could be due to the ineffectiveness of the FFT in extracting information from short duration non-cyclical signals or the low sample rate employed when compared to that used by snickometer (44,100 kHz [10]). This Figure additionally shows that SVMs were the most accurate type of algorithm for classifying impacts from the 16-27 feature set. This suggests that these features extracted from the accelerometer FFTs can be separated accurately by different hyperplanes. Figure V-D shows a plot of the features 16-27 from Table II per class of the entire dataset, with 1 signifying a hit, 2 signifying an edge and 3 signifying a miss. This shows the most distinctive features are 21 and 25 from Table II in classifying impacts. These features are the accelerometer FFT covariance of the Y-axis with respect to the X and Z axes and confirm the theory suggested in Subsection V-B.

E. Algorithm Selection

The most accurate ML algorithm was found using the most accurate features identified in Subsection V-D. The same dataset and validation procedure were used as in Section V-D, however, due to small variations in the accuracy each run caused by the randomisation within the validation procedure, each algorithm in Table I was tested three times and an average computed. The results can be seen in Figure 14. The top performing algorithm was the Linear SVM with an average accuracy of 88.2% and a peak accuracy of 89.3%. The other top performing algorithms in order of average accuracy were Subspace Discriminant (87.8%), Quadratic SVM (86.5%), Medium Gaussian SVM (86.1%) and Weighted KNN (85.7%). Within the top 5 performing algorithms there are 3 different

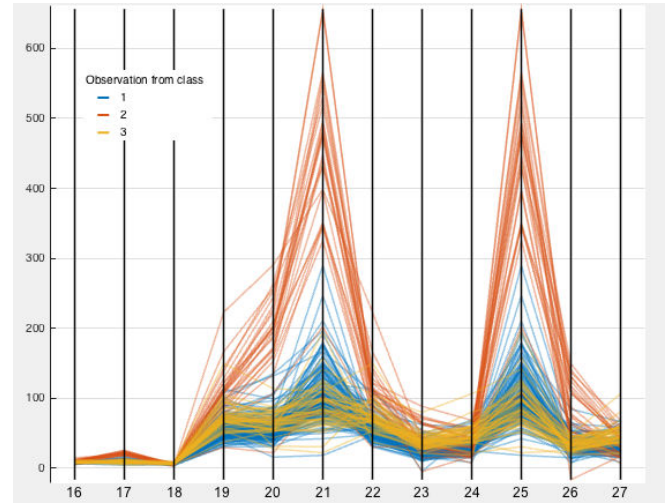


Fig. 13. Plot of the most accurate feature set (16-27 from Table II) per class

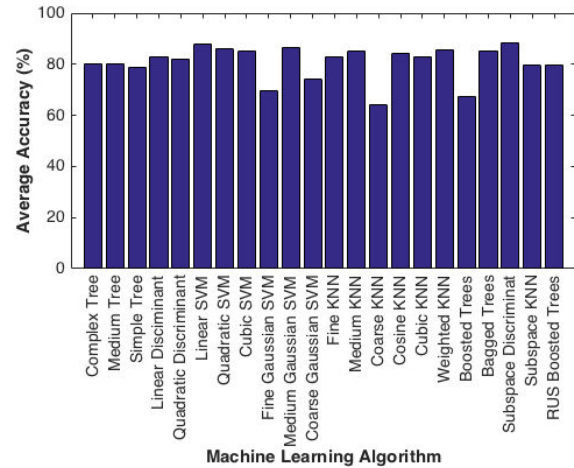


Fig. 14. Average accuracy across three runs of each ML algorithm for the most accurate feature set (16-27 from Table II)

types of ML algorithms. This result shows the features chosen classify impacts accurately using different ML methodologies.

To analyse the performance of the best performing algorithm further, a confusion matrix per true class for the linear SVM with features 16-27 and an overall accuracy of 89.3% can be seen in Figure 15. Class 1 represents a hit, class 2 represents an edge and class 3 represents a miss respectively. This diagram shows that the most accurate classification was hit with 94.7% accuracy, however this was to be expected with the largest dataset of hits. The aim of this research was to identify edges, as these are the most difficult to detect for umpires in making caught decisions. This Figure clearly shows no confusion between misses and edges. As in the eyes of the umpire edges and hits result in the same decision, this is a significant result.

F. Sample Size

Another aim of this research was to assess whether the accuracy of ML algorithms improve as the dataset increases in

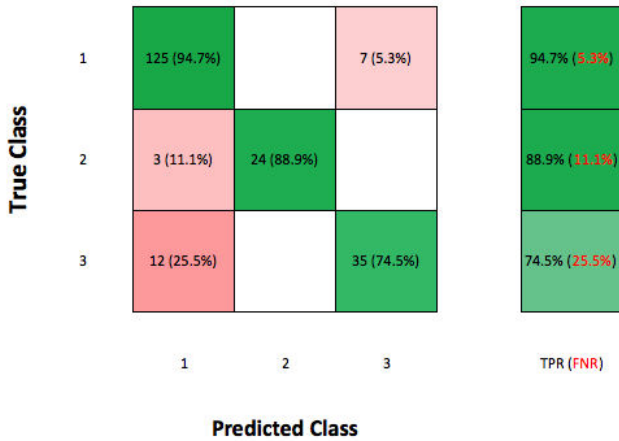


Fig. 15. Confusion matrix for the linear SVM across the entire dataset (overall accuracy of 89.3%) [TPR: True Positive Rate, FNR: False Negative Rate]

size. In order to test this, the dataset was randomly split into three equal subsets. The small dataset contains 68 samples (44 hits, 9 edges and 15 misses), the medium dataset contains 136 samples (88 hits, 18 edges and 30 misses) and the large dataset contains 204 samples (132 hits, 27 edges and 45 misses). The top five performing algorithms from Subsection V-E were tested on the datasets of varying size, utilising the same five fold validation procedure throughout.

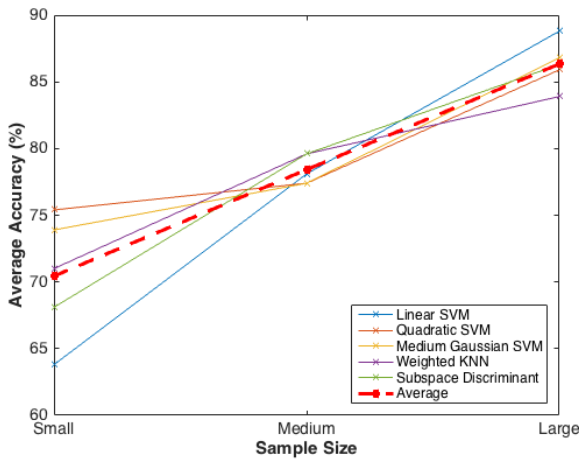


Fig. 16. Average accuracy of the top 5 performing ML algorithms as dataset increases in size

Figure 16 shows the accuracy of the algorithms as the dataset increases in size. From the results, it is clear that all the algorithms accuracy improve as the dataset increases in size. However, from the calculation of the standard deviation of the accuracies, the algorithms do not become more consistent. This is contrary to the findings of Banko and Brill [6] and could be caused by the comparatively small datasets used in this experimentation compared to that study.

VI. FURTHER WORK

Despite a positive result in classifying impacts between a cricket bat and ball through the methods presented, there are many areas to expand the research into. The current procedure does not consider a number of factors that are involved when making a caught decision. For example, there are a number of scenarios that may effect the vibrations of the bat such as the bat hitting the pad or ground, but not the ball, that are not factored into the experimentation. There are also other classes of impact that are ruled as a hit that are not considered, such as the ball hitting the batsman's gloves. Further research into the classification of these additional impacts would enable the device to assist umpires further. In addition to this, future research would be required to expand the dataset to different cricket shots, as the current procedure only considers an off drive. A significant amount of additional data would be required to classify other shots such as the pull, hook and cut accurately.

One factor that may have impacted the results of the research presented is the uneven distribution of the dataset. This was due to the unsuitability of the components to the application, restricting data collection. Furthermore, it was much more difficult to get edges than any other class of impact. This is because a batsman never intends to edge a cricket ball and therefore it occurs less regularly than a hit or miss that can be performed on demand. Additional research would enable an evenly distributed dataset to be collected, to assess the impact on the accuracy of the algorithms.

Finally, before being used in a game situation, the design of the device and components would have to be optimised significantly to minimize the impact on the weight and appearance of the bat. Alongside this a method of processing the data obtained on the device and transferring the decision from the bat to the umpire would have to be devised.

VII. CONCLUSION

In this paper, a device has been designed to be attached to the back of a cricket bat, to measure acceleration and sound created during impact of multiple off drives. The data from this device has been processed and experimentation conducted in order to find the most accurate combination of features and ML algorithms for classifying the impacts as hits, edges or misses. The most accurate combination of features for classifying impacts was found to be the covariance and standard deviation of the FFT of the post impact accelerometer data (16-27 in Table II). Within this feature set in particular, the covariance of the Y-axis with respect to the X and Z axes were significant features. The most accurate type of ML algorithm, with this combination of features, was found to be support vector machines (SVMs) with an average accuracy of 84%. A linear support vector machine was found to be the most accurate algorithm with an average accuracy of 87.8% and a peak accuracy of 89.3%. In addition, the confusion matrix for the linear SVM with 89.3% accuracy shows that there is no confusion between edges and misses. As edges and hits are classified as the same by umpires when making caught decisions, the device detected and classified all 27 of the

edges correctly from a decision-making point of view. The peak accuracy of 89.3% for the linear SVM in classifying impacts is comparable to the ICC stated accuracy of 91% for professional umpires. Furthermore, the accuracy of the top performing classifiers all increase with dataset size, suggesting that the accuracy of the ML algorithms could exceed the ICC requirement with more data. However, a number of factors when making caught decisions have not been taken into account in this experimentation. Therefore, further research would be required before the device was ready to use in a game situation.

ACKNOWLEDGMENT

The author would like to thank Andrew Gallant for his support and guidance whilst supervising the research project; Paul Grayson and DUCC for allowing testing of the device when required; Ian Hutchinson and Neil Clary for their assistance designing, building and repairing the device; Ben Fothergill for his effort and time as a test subject and Richard O'Dair for countless hours spent coaching the author.

REFERENCES

- [1] V. K. Borooah, "Upstairs and downstairs the imperfections of cricket's decision review system," *Journal of Sports Economics*, vol. 17, no. 1, pp. 64–85, 2016.
- [2] A. Busch and D. James, "Analysis of cricket shots using inertial sensors," *The impact of technology on sport II*, pp. 317–322, 2007.
- [3] L. Fallon, J. Sherwood, and M. Donaruma, *An Assessment of Sensing Technologies to Monitor The Collision of a Baseball and Bat (P34)*. Paris: Springer Paris, 2008, pp. 191–198. [Online]. Available: http://dx.doi.org/10.1007/978-2-287-09411-8_23
- [4] S. Ting and M. Chilukuri, "Novel pattern recognition technique for an intelligent cricket decision making system," in *Instrumentation and Measurement Technology Conference, 2009. I2MTC'09. IEEE*. IEEE, 2009, pp. 978–982.
- [5] T. M. Mitchell *et al.*, "Machine learning. web," 1997.
- [6] M. Banko and E. Brill, "Scaling to very very large corpora for natural language disambiguation," in *Proceedings of the 39th Annual Meeting on Association for Computational Linguistics*, ser. ACL '01. Stroudsburg, PA, USA: Association for Computational Linguistics, 2001, pp. 26–33. [Online]. Available: <http://dx.doi.org/10.3115/1073012.1073017>
- [7] R. Steen, "Going upstairs: The decision review system—velvet revolution or thin edge of an ethical wedge?" *Sport in Society*, vol. 14, no. 10, pp. 1428–1440, 2011.
- [8] R. Cross, "The sweet spots of a tennis racquet," *Sports Engineering*, vol. 1, no. 2, pp. 63–78, 1998.
- [9] A. K. Sarkar, D. A. James, A. W. Busch, and D. V. Thiel, "Cricket bat acceleration profile from sweet-spot impacts," *Procedia Engineering*, vol. 34, pp. 467 – 472, 2012. [Online]. Available: <http://www.sciencedirect.com/science/article/pii/S1877705812016931>
- [10] R. Rock, A. Als, P. Gibbs, and C. Hunte, "The 5th umpire: automating crickets edge detection system," *Journal of Education, Informatics & Cybernetics*, vol. 11, no. 1, 2013.
- [11] BBC, "<http://www.bbc.co.uk/sport/cricket/36102233>," April 2016.
- [12] S. Cole, "Has computer technology enhanced international cricket?" *Computers for Everyone*, p. 42.
- [13] TopEndSports, "Cricket technology," January 2016. [Online]. Available: <http://www.topendsports.com/sport/cricket/technology.htm>
- [14] N. Egan, "Bat for ashes: Technology used at the cricket," March 2014.
- [15] V. Mali, "<https://www.globalcricketcommunity.com/index.php/396-devices-and-technologies-used-by-umpires-during-a-cricket-match>," May 2016.
- [16] Hawkeye, "<http://www.hawkeyeinnovations.co.uk/sports/cricket>," March 2017.
- [17] R. Shivakumar, "What technology says about decision-making: Evidence from cricket's decision review system (drs)," *Available at SSRN 2700405*, 2015.
- [18] H. Collins and R. Evans, "Sport-decision aids and the "csi-effect": why cricket uses hawk-eye well and tennis uses it badly," *Public Understanding of Science*, vol. 21, no. 8, pp. 904–921, 2012.
- [19] A. Narayanan, "Cricket dismissals through the ages," ESPN cricinfo, January 2013.
- [20] L. Rokach and O. Maimon, *Data mining with decision trees: theory and applications*. World scientific, 2014.
- [21] J. R. Quinlan, "Induction of decision trees," *Machine learning*, vol. 1, no. 1, pp. 81–106, 1986.
- [22] R. A. Fisher, "The use of multiple measurements in taxonomic problems," *Annals of eugenics*, vol. 7, no. 2, pp. 179–188, 1936.
- [23] G. McLachlan, *Discriminant analysis and statistical pattern recognition*. John Wiley & Sons, 2004, vol. 544.
- [24] T. Zhang, "An introduction to support vector machines and other kernel-based learning methods," *AI Magazine*, vol. 22, no. 2, p. 103, 2001.
- [25] T.-M. Huang, V. Kecman, and I. Kopriva, *Kernel based algorithms for mining huge data sets*. Springer, 2006, vol. 1.
- [26] N. S. Altman, "An introduction to kernel and nearest-neighbor non-parametric regression," *The American Statistician*, vol. 46, no. 3, pp. 175–185, 1992.
- [27] D. Coomans and D. L. Massart, "Alternative k-nearest neighbour rules in supervised pattern recognition: Part 1. k-nearest neighbour classification by using alternative voting rules," *Analytica Chimica Acta*, vol. 136, pp. 15–27, 1982.
- [28] R. Polikar, "Ensemble based systems in decision making," *IEEE Circuits and systems magazine*, vol. 6, no. 3, pp. 21–45, 2006.
- [29] L. Rokach, "Ensemble-based classifiers," *Artificial Intelligence Review*, vol. 33, no. 1-2, pp. 1–39, 2010.
- [30] Mathworks, "Classification learner app," March 2017. [Online]. Available: <https://www.mathworks.com/help/stats/classificationlearner-app.html>
- [31] J. Schneider, "<https://www.cs.cmu.edu/schneide/tut5/node42.html>," February 1997.
- [32] ARM, "<https://developer.mbed.org/platforms/mbed-lpc1768/>," February 2017.
- [33] GP, "<http://www.gpbatteries.com/inten/portable-powerbank/everyday/511pb>," February 2017.
- [34] Analog, "<http://www.analog.com/en/products/accelerometers/adxl362>," February 2017.
- [35] Sparkfun, "<https://www.sparkfun.com/products/9868>," February 2017.
- [36] BOLA, "<http://www.bola.co.uk/productlist.html>," March 2017.
- [37] G. Bunkheila, "Signal processing and machine learning techniques for sensor data analytics," Webinar, September 2015.
- [38] D. P. Morgan and C. L. Scofield, "Signal processing and feature extraction," in *Neural Networks and Speech Processing*. Springer, 1991, pp. 163–201.